

The protocol of Western blotting

1. SDS-PAGE

1. **Design your western blot for Pull Down assay before class.** All the samples should be treated with 2 × sample buffer (loading buffer) for SDS-PAGE with the ratio of 1:1 (v/v).
 - Three samples used in this experiment: A: Strep-sumo-A; C: Strep-sumo; B: T7-tag-B.
 - Think about:
 - (1) how to test Strep-tag proteins (A and C)?
 - (2) how to test input (T7-tag-B protein)?
 - (3) how to examine protein-protein interaction?
 - (4) what control should be considered in your design?
 - You have samples: Pull down sample-beads in tube AB; Pull down sample-beads in tube CB.
 - You can use prepared input sample directly (prepared by TA or instructor).
 - Primary antibody will be provided: mouse α Strep (1:5000); mouse α T7 (1:5000).
 - Secondary antibody will be provided: Goat-anti-mouse-IgG(H/L)-AP (or HRP)
2. Prepare a SDS-Polyacrylamide Gel (12%) to run samples.
3. Load samples based on your design;
4. Run the gel to separate proteins;
5. Take the gel from gel box (cut it based on your design if necessary);
6. Stain part of the gel with Coomassie Brilliant Blue R250 overnight if it is necessary for your result analysis. Destain it later. [Or you can run two gels; one is for Coomassie Brilliant Blue stain and the other one is for transferring.]

2. Transferring

7. Prepare 8 copies of filter paper and one nitrocellulose membrane in the same size as the gel; soak filter paper in transfer buffer for at least 5-10 min, be sure to drive off the bubbles between them; rinse the nitrocellulose membrane before use;
8. Place the gel, paper, and membrane into the blotting tank in the following order (Fig.1);



Fig.1 The Order of Transfer Electrophoresis

9. Close the tank, then make constant-current transferring electrophoresis ($I = \text{Gel surface area } A/\text{cm}^2 \times 0.8 \text{ mA}$) for 40-60 min; after transferring, rinse the membrane by TBS for

3-5 s.

10. The membrane is soaked in block solution (2-3% non-fat-dry milk in TBST) to block non-specific protein-binding sites at 4 °C overnight. (Blocking also can be performed at room temperature for 1-2 h according to optimized protocol.)

3. Immunoblotting

11. Wash the membrane with fresh TBST and keep in liquid reagent to prevent drying in following steps;
12. Cover the membrane with about 5 mL the first antibody (1:8000-1:5000 diluted in TBST) into a plate (Primary antibody is Mouse-anti-X protein, shake the plate at RT for 1.5 h);
13. Wash the membrane with fresh TBST 3×5 min to remove unbound antibodies;
14. Cover the membrane with about 5 mL secondary antibody solution [Goat-anti-mouse-IgG(H/L)-AP (alkaline phosphatase)(or-HRP, Horseradish Peroxidase), 1:5000 diluted in TBST] into a new plate, shake the plate at RT for 30 min;
15. Wash the membrane with TBST 3×5 min;
16. Wash with TBS for 5 s to remove Tween-20.
17. Develop the membrane in 10 mL substrate solution.
18. Stop the developing process in ddH₂O and dry the membrane in filter paper directly. (The color of bands would fade in a few days.)

Appendix

1. Transfer Buffer: 25mmol/L Tris, 200mmol/L Glycine, 20% Methanol, 0.1%SDS, pH 8.5;
2. TBS (20mmol/L Tris-HCl, 150 mmol/L NaCl, pH7.5);
3. TBST (20mmol/L Tris-HCl, 150 mmol/L NaCl, pH7.5, 0.05%-v/v Tween 20);
4. Block solution: 2-3% non-fat-dry milk in TBST;
5. Substrate solution for **AP** (Alkaline Phosphatase):
Buffer: 100 mmol/L Tris/HCl, 100 mmol/L NaCl, 5 mmol/L MgCl₂ pH9.5.
Add 35 µL BCIP-T (5-bromo-4-chloro-3-indolyl phosphate, p-toluidine salt, 50 mg/mL in Dimethyl formamide) and 45 µL of NBT (Nitro blue Tetrazolium, 50mg/mL in Dimethyl formamide) to 10 mL of buffer for development.
6. Substrate solution for **HRP** (Horseradish Peroxidase):
A) Dissolve 30 mg CN(4-chloro-1- naphthol, Sigma) in 5 mL methaonl;
B) Dissolve 10 mg DAB (3,3'-diaminobenzidine tetrahydro-chloride, Sigma) in 5 mL methanol;
C) Stir A and B separately for 10-15min to dissolve completely, then mix A and B, add PBS to 50 mL, and aliquot and freeze the unused portions;
D) Take 10 mL fresh solution, add 10 µL 30% H₂O₂ when using.
7. Substrate solution for **HRP** (Horseradish Peroxidase):
ECL Chemiluminescent Substrate Detection Kit